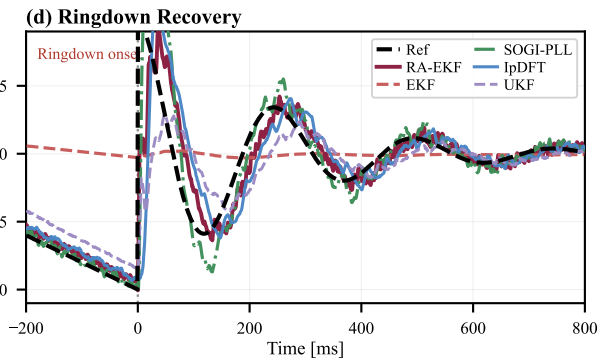
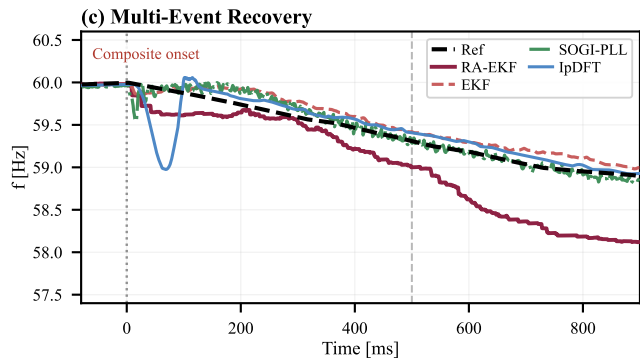
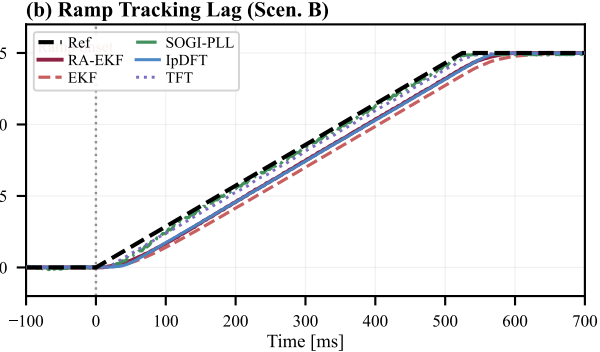
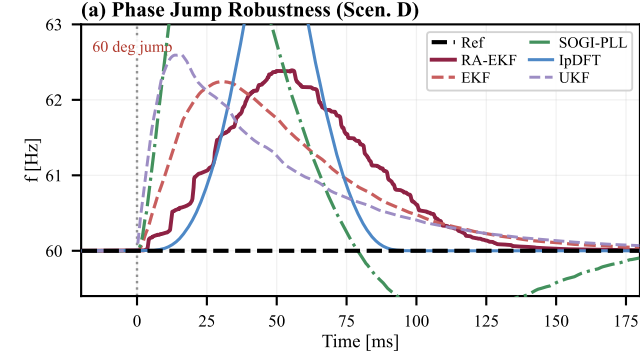




**(e) Disturbance Scorecard**

|          | Step | Ramp | Mod | Ring | Multi |
|----------|------|------|-----|------|-------|
| Koopman  | F    | P    | P   | F    | F     |
| TFT      | P    | P    | P   | F    | F     |
| IpDFT    | F    | F    | F   | F    | F     |
| SOGI-PLL | F    | P*   | F   | F    | F     |
| SOGI-FLL | F    | F    | F   | P    | F     |
| PLL      | F    | F    | F   | F    | P*    |
| EKF      | P*   | F    | F   | P    | P     |
| UKF      | F    | F    | F   | P*   | P     |
| RA-EKF   | P    | F    | P*  | F    | F     |

P = top-3 RMSE within family, P\* = best

